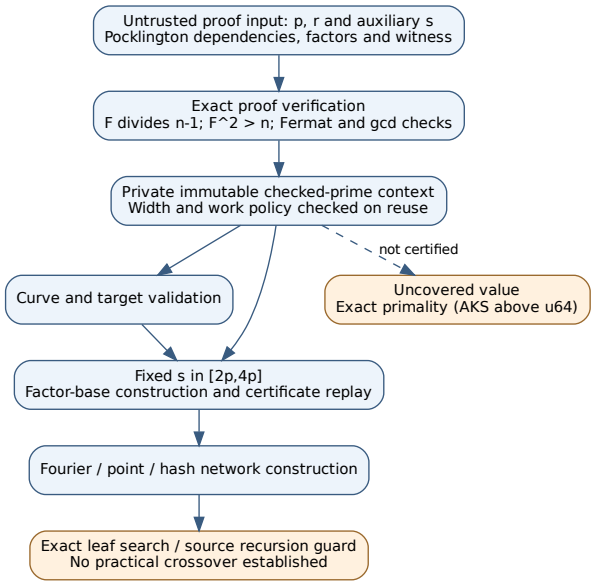
Checked hash-prime setup for native ECCp-131

Engineering and bounded correctness receipt

30 October 2026

The curve already accepts checked p/r proofs. This follow-up retains verified auxiliary prime facts for hash-modulus setup, factor bases and replay. The source permits a fixed prime $s \in [2p, 4p]$; uniform hash seeds are preserved. An optional declared modulus enables this path. Default random selection keeps its candidate stream and RNG draws. Uncovered values retain exact primality.



Development evidence. Six targeted controls pass, including a 65-bit proof, official 132-bit hash modulus, resource limits, missing proof, tiny known-answer solve/replay and runner pin rejection. Official parameter, target and hash preflight charges 5,897 uncalibrated steps under the original two-million-step policy. Fixture preparation checks 68 candidate proofs and charges 371,334 diagnostic steps. These are not timing measurements; proof generation belongs in future cold accounting. No runtime gain is established.

Limits. The unchanged literal recursion guard requires $E > 2^{131070}$ before its root scale can become positive; additional guards also apply. Machine-sized graph indices keep this production scale zero. Generic uncertified large-prime setup, practical search, both challenge solves and the independent full audit remain open. No profile or scientific benchmark ran. The user's Fourier-construction timings are attributed observations, not independently admitted results here; primality hangs do not show construction infeasibility. S3 checkpoint upload/readback has succeeded separately.

Sources and evidence. Bochum v0.2, section 3; HAC Fact 4.40; the existing native source guards, new checked JSON fixture and retained development transcripts/source manifests beside README.md. Existing frozen reports, theorem constants, curves and leaderboard evidence are unchanged.